Prabhavit Gupta

Senior AI Engineer | Generative AI, LLMs, RAG Pipelines, Multi-Agent Systems & MLOps

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Executive Summary

- **Generative AI & Multi-Agent Architecture Design:** Applied AI engineer with 3 years commercial software development experience designing and deploying enterprise Generative AI systems. Proven expertise constructing advanced LLM solutions, multi-agent frameworks, dynamic tool routing, and automated fallback logic across production environments.
- **End-to-End RAG Pipelines & Knowledge Retrieval:** Expert in developing high-throughput Retrieval-Augmented Generation (RAG) pipelines over heterogeneous unstructured document repositories. Proficient in semantic chunking, dense vector embeddings, reranking strategies, and database optimization with PostgreSQL pgvector and Qdrant.
- **Production Observability, Evaluation & MLOps:** Dedicated to industrializing AI solutions with end-to-end traceability, structured logging, hallucination mitigation guardrails, and automated regression evaluation suites (RAGAS framework), ensuring sustained reliability and regulatory compliance.

Technical Skills

- **Generative AI & LLM Engineering:** OpenAI (GPT-4o, o1), Anthropic Claude, Open-Weight Models (Llama, Mistral), Prompt Engineering, ReAct Planning, Tool Calling, Structured Outputs (Pydantic / JSON Schema).
- **Agentic Architectures & Frameworks:** LangGraph, LangChain, LlamaIndex, Multi-Agent Systems, State Graphs, Memory Management, Human-in-the-Loop Escalation.
- **RAG, Vector Stores & Search:** Retrieval-Augmented Generation (RAG), Document Ingestion, Semantic Chunking, PostgreSQL pgvector, Qdrant, ChromaDB, Hybrid Search, Cohere Rerank.
- **Backend Engineering & DevOps:** Python (AsyncIO, FastAPI, PyTest), TypeScript, Node.js, REST/gRPC APIs, Docker, Git, GitHub Actions CI/CD, AWS & Cloud Environments.
- **Evaluation, Observability & Safety:** RAGAS Benchmarking, Precision/Recall, Hallucination Prevention, Guardrails against Prompt Injection, OpenTelemetry Tracing, Structured Logging.

Professional Experience

Senior AI Engineer & Solutions Consultant • *Independent / AI Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Designed, developed, and industrialized customized Generative AI and multi-agent solutions for enterprise clients, translating complex operational requirements into scalable software architectures.
- Built production RAG pipelines over complex document archives using PostgreSQL pgvector and Qdrant, implementing semantic chunking and reranking to achieve sub-second query response times with zero hallucination.
- Orchestrated collaborative multi-agent workflows using LangGraph and Python, managing deterministic state transitions, automated retry mechanisms, and human-in-the-loop validation checkpoints.
- Established automated evaluation pipelines using the RAGAS framework to benchmark model outputs, context retrieval fidelity, and latency across iterative prompt and model upgrades.
- Containerized services with Docker and implemented continuous integration/deployment (CI/CD) pipelines, incorporating structured logging and distributed tracing for production observability.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed asynchronous microservices in Python and TypeScript for high-volume document parsing, entity extraction, and automated categorization.
- Architected vector indexing jobs and background data ingestion queues using Redis and PostgreSQL, ensuring scalable data processing and operational resilience.
- Created comprehensive unit and integration test suites using PyTest, maintaining high code quality and test coverage across iterative releases.
- Collaborated closely with domain specialists and product owners to design user-centric AI workflows and evaluate system performance against business KPIs.

Key Engineering Projects

- **Enterprise Multi-Agent Orchestration Platform (LangGraph + Python):** Multi-agent collaborative pipeline coordinating document classification, extraction, synthesis, and verification agents with automated state preservation and audit logging.
- **Production Knowledge Retrieval Engine (Python + pgvector):** Industrialized RAG system combining dense vector search, hybrid keyword retrieval, semantic caching, and automated hallucination guardrails over complex

institutional records.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, decision analysis, analytical problem solving, and complex system modeling.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency), Spanish (Elementary reading/comprehension).
- **Legal Authorization:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union without visa sponsorship.